



Towards a framework for local interrogation of AI ethics: A case study on text generators, academic integrity, and composing with ChatGPT

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ABSTRACT

Ethical frameworks for text generators (TGs) in education are generally concerned with personalized instruction, a dependency on data, biases in training data, academic integrity, and lack of creativity from students. While broad-level, institutional guidelines provide value in understanding the ethical dimensions of artificial intelligence (AI) for the classroom, there is a need for a more ecological understanding of how AI ethics might be constructed locally, one that takes into account the negotiation of AI between teacher and student. This article investigates how an educational ethical framework for AI use emerges through a qualitative case study of one composition student's interaction with and understanding of using ChatGPT as a type of writing partner. Analysis of interview data and student logs uncover what we term an emergent "local ethic" – a framework that is capable of exploring unique ethical considerations, values, and norms that develop at the most foundational unit of higher education – the individual classroom. Our framework is meant to provide a heuristic for other writing teacher-scholars as they interrogate issues related to pedagogy, student criticality, agency, reliability, and access within the context of powerful AI systems.

Introduction

With the arrival of OpenAI's ChatGPT and other artificial intelligence (AI) text generators (TGs), writing studies finds itself at a critical inflection point concerning policy related to academic integrity, assessment, and access. While research across fields on AI literacy is still nascent, a growing body of work in communication, tech policy, and computer science has led to insights involving the tension between legal and ethical action (Wagner, 2019) and how the deployment of ethical frameworks for AI governance lack the mechanisms for compliance (Larsson, 2020). Scholars have started to track the recent uptake of regulatory initiatives on AI across the globe but particularly in Europe, Northern America, and Asia (Daly et al., 2019; Niklas & Lina, 2020), analyzing regulatory measures and investments while foregrounding ethics as a normative framework (Jobin et al., 2019). While this rise of ethical guidelines may reflect a strategic move by the corporate sector to escape actual regulation (Wagner, 2018), it also informs educational tools and policies for governance (Larsson, 2020).

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In the educational space, ethical frameworks are so far concerned with personalized instruction, a dependency on data, biases in training data, academic integrity, and lack of creativity from students (Baidoo-Anu & Owusu Ansah, 2023; Bolukbasi et al., 2016). Open AI's recent guidebook for instructors and the U.S. Department of Education have considered broader frameworks, while the MLA/CCCC Joint Task Force on Writing and AI has provided pointed recommendations. These guidelines provide important value in understanding the ethical dimensions of AI for the classroom. However, there is a need for a more ecological understanding of how AI ethics might emerge, one that takes into account the negotiation of AI on the ground level between teacher and student. This article works towards that aim by investigating how an educational ethical framework for AI use emerges through a case study of one composition student's interaction and understanding of using ChatGPT.

By situating our local investigation within larger discussions of policy and governance, we hope not only to uncover some of the most common missteps in AI policy and resulting (in)compliance, but also make connections between ethical AI discourses at both broad public policy levels as well as more localized instruction (e.g. the college classroom). This article analyzes interview data and student logs to uncover what we term an emergent "local ethic" – a framework that is capable of exploring unique ethical considerations, values, and norms that develop at the most foundational unit of higher education – the individual classroom. Ultimately, our framework is meant to provide a heuristic for other writing teacher-scholars as they interrogate issues related to pedagogy, student criticality, agency, reliability, and access within the context of powerful AI systems.

Our central argument, based on these findings and review of the literature, is simple yet significant. First, writing educators need to assume that at least a portion of their students will experiment with and adapt AI-powered TG tools. Second, whether we explicitly address the use of these tools in our classrooms or not, we are actively participating in the creation of a local ethic governing the use of AI. To better position both ourselves and our students as teachers and learners of writing, we must actively engage in explorations of AI ethics. Accordingly, our contribution to this special issue highlights the importance of exploring local rhetorical situations, while also ensuring alignment with overarching policies to maintain consistency and fairness across courses and classrooms. In the evolving landscape of AI integration, our article encourages an ongoing dialogue that considers the dynamic interrelationship between local ethical considerations and global established guidelines such as the MLA-CCCC recommendations, fostering a comprehensive and adaptable approach to AI ethics in writing instruction.

Ethical guidelines surrounding AI

Various stakeholders have tried to define broad policy guidelines for AI implementation across disciplines, industries and economic sectors. Many of these stakeholders have emphasized the need for accountability for AI systems (Bender, Gebru, McMillan-Major, & Shmitchell, 2021; Campolo & Crawford, 2020; Chopra & Singh, 2021), or the desire for natural language processing (NLP) models to be used as a supplement, not as a replacement, for human interaction (Fuchs, 2023). From a theoretical perspective, moral ethicists have considered "whether AI should make moral decisions like a consequentialist, a deontologist, a virtue theorist, or more simply on a case-by-case basis" (Pflanzer et al., 2023, p. 1268). Scholars have also wondered if AI can be morally responsible at all. Some remain deeply skeptical (Sison & Redín, 2021; Sparrow, 2021), while others have argued that AI might already have the reactive attitudes necessary to be considered morally responsible (Maruyama, 2022). It's important to note, however, that contemporary arguments on moral responsibility are still seeking out a unified set of guidelines to follow in order to be morally responsible, and recent scholars have wondered if such guidelines can be articulated given the range of approaches and conditions when it comes to moral responsibility (Smilansky, 2000; Strawson, 1994).

Recently, a various set of ethical approaches have been articulated regarding AI tools. Experts have argued that ethical guidelines should minimize the potential inherent bias within large language model (LLM) systems (Fuchs, 2023; Zhou et al., 2023), and should require legal and social experts to explore ethical guidelines for applications such as ChatGPT. Calls for ethical guidelines continue to strike a balance between healthy and excessive use of AI, noting that stakeholders should avoid over-regulating which could lead to stifling innovation, but promote the healthy discussion between emerging technologies and moral behavior (Zhou et al., 2023). This sense of balance can be seen in the recent law passed by the European Union entitled the A.I Act. Arguably the world's first comprehensive attempt to limit AI, the act focuses on a "risk-based approach" that requires companies that pose the most potential harm to society to provide a risk assessment to regulators regarding their AI tools (Satariano, 2023, n.p.).

A more nuanced approach to AI ethics has also been issued in higher education discussions. Scholars have recognized the benefits of ChatGPT while acknowledging its effects on academic integrity. These effects include, but are not limited to, how students experience the transformative aspect of education, manage the meaning-making process and discover novel ideas in research (Robin, 2023). Within professional communications, Alafnan et al. (2023) examined ChatGPT usage by conducting 30 theory-based and application-based tests. They found that ChatGPT has the potential of replacing search engines as it provides accurate and reliable input to students. Their recommendations, which were fairly specific, included refraining from making theory-based questions as take-home assessments, providing business writing students with detailed scenario-based assessment tasks that call for personalized answers, and utilizing critical and creative thinking. They also suggested integrating and reviewing ChatGPT-generated responses in class. Adjusting the classroom experience has also been echoed by ChatGPT's own creators, Open AI. They assert that ChatGPT can help accelerate learning by having AI operate as a stand-in for a particular persona, assist in generating learning materials like tests and quizzes, and educate students about critical thinking (Teaching with AI, 2023). The U.S. Department of Education (DoE) considers recommendations more broadly with their recently released *Artificial Intelligence and the Future of Teaching and Learning: Insights and Recommendations* (Cardona et al., 2023a). While explicating the opportunities and potential benefits of AI for "enabl[ing] new forms of interaction," "help[ing] educators address variability in student learning," "support[ing] powerful forms of adaptivity," "enhanc[ing] feedback loops," and "support[ing] educators," the DoE recognizes that AI will "increase existing risks and introduce new risks"

especially related to data privacy and security (Cordona et al., 2023b, p. 2). The main thrust of the document provides an introduction of “four key foundations” as a “starting point” for framing future policy creation and adoption: “Foundation 1: Center People (Parents, Educators, and Students); Foundation 2: Advance Equity; Foundation 3: Ensure Safety, Ethics, and Effectiveness; Foundation 4: Promote Transparency” (Cardona et al., 2023a, pp. 6–9).

More specific to English studies, the Modern Language Association (MLA) and the Conference on College Composition and Communication (CCCC) created a joint task force to develop guidelines and professional standards around the use of AI and writing (MLA-CCC Joint Task Force, 2023). Unlike AlAfnan et al.’s focus on student usage, their working paper outlines the risks and benefits to AI in the writing classroom for both students and teachers. Their broad recommendations include providing support for teachers that responds to the complexity of the issue and labor involved, and centering the continued teaching of writing as a mode of learning, exploration, and literacy development. They also emphasize academic integrity policies that support rather than punish students’ usage of AI and avoid restrictions in the writing classroom that make writing for the class radically different than writing in other environments (MLA-CCC Joint Task Force, 2023). These discussions regarding AI ethics account for multiple stakeholders while trying to balance AI’s valuable contributions and possible drawbacks.

Ethics of AI in writing studies

In recent years, writing studies scholarship has undergone a significant shift in its approach to ethics writ large, which involves a reevaluation that emphasizes the development and negotiation of ethical dispositions within specific rhetorical contexts. In discussions about the relationship between ethics and writing instruction, there is a tendency to rely on characterological measures of ethical behavior (Duffy, 2017; Frisicario-Pawlowski, 2018; Helmers, 1994). These measures involve idealized expressions of personal integrity or public morality that serve as the basis for ethical commitments. As such, historically, the development of models for ethics instruction in composition often emphasizes ethical outcomes in terms of personal character. Helmers (1994) critiques that in composition testimonials, student writing is frequently linked to perceptions of an individual’s moral codes or principles. Despite efforts in composition research to separate perceptions of error from judgments of character, the persistent use of characterological measures in learning discussions may perpetuate this pattern. In response to the overemphasis on personal character that is oftentimes static, Duffy (2017) insists that “virtues are context-dependent, responsive to the *kairetic* moment, and social in nature, expressing the values, traditions, and narratives of specific communities and cultures” (p. 235). Put otherwise, scholars have increasingly called for a rethinking of ethics, moving toward a recognition of the everchanging rhetorical situations within local contexts.

This shift in thinking about ethics also parallels other theoretical advances in digital writing and rhetoric, such as a rethinking of agency and authorship as distributed and dispersed (Cooper, 2019; Gries, 2015; Jiang, 2019; Jiang & Tham, 2022; Jordan, 2015; Micciche, 2014). Specifically, scholars have cast doubt on the explicit teaching of universal ethical values within writing classrooms (Duffy, 2017; Frisicario-Pawlowski, 2018; Kraemer, 2017), along with the prevailing emphasis on value-based instruction in writing studies that prioritizes the indoctrination of fixed values. Kraemer (2017) observes that, rather than imparting universally applicable values, it is more crucial for students to adapt their actions to changing rhetorical situations. This developmental perspective on ethical instruction is also endorsed by Frisicario-Pawlowski (2018), who presents a dialogical model of ethics that revolves around the contextual nature of ethical behaviors. Frisicario-Pawlowski (2018) emphasizes that ethical actions should be understood as being “situated in local rhetorical contexts” (p. 124). In essence, instead of promoting fixed virtue ethics, writing scholars and teachers should guide students to perceive ethics as an ongoing dialogue, constantly negotiated across various rhetorical contexts and ecologies. In much the same way, there is not a single universal ethic for AI usage and we do not seek to create or gesture toward such, but rather move the field towards a framework for local negotiation and investigation, one that is also responsive to institutional and disciplinary guidelines.

This dialogical understanding of ethics, as intertwined with social, material, and rhetorical contexts, necessitates examinations of AI authoring tools, including TGs, within specific localities. This repositioning occurs not primarily through the imposition of values, but through the construction of distinct rhetorical ecologies that reflect everchanging networked relations. In order to frame a better understanding of the rhetorical ecology (Rivers & Weber, 2011) regarding AI and human-machine interactions, writing studies scholars (Beveridge et al., 2020; Gallagher, 2017; Jiang & Vetter, 2020; Tham et al., 2022) have started to more progressively incorporate concepts from new materialist rhetorics (Barnett & Boyle, 2016; Gries, 2015), posthumanist rhetoric (Boyle, 2018; Dobrin, 2015), actor-network theory (Latour, 2005), and object-oriented ontology (Bogost, 2012; Harman, 2018). These theoretical perspectives highlight the significance of the interconnected and dynamic interactions between humans and nonhumans and have implications for a number of ethical issues. Most notably, agency is not confined to individual entities but is distributed across various actors. Together, humans and nonhumans collectively create agential capacities that continually adapt and reshape in reaction to ongoing movement and socio-material action. For instance, AI-powered TGs introduce a collaborative element, as students work in tandem with AI tools to co-create written content. The agency of writing becomes distributed across both human and nonhuman actors, generating a synergistic effect. In this sense, AI TGs challenge conventional notions of authorship and illuminate the potential of technology as a co-agent in the writing process.

Furthermore, human-machine relations within rhetorical ecologies are not static but continually reconfigured in response to various forces and actions. Echoing writing studies scholars’ reevaluation of ethics as fluid and situated, Colton and Holmes (2018) similarly maintain that virtue ethics should not be construed as “fixed, inflexible, and privileging only those in power” (p. 35). Crucially, they argue for a departure from relying solely on a universal axiom for classifying ethical and social justice practices, acknowledging the adaptability and fluidity inherent in the evolution of ethical concepts. Colton and Holmes (2018) have repurposed Latour’s (2004) ethical theory concerning fairness, merging it with Brown’s (2015) notion of hospitality in *Ethical Programs*, to outline

potential paths for ethical considerations within interconnected societies and digital rhetorical contexts. This reinterpreted sense of fairness paves the way for establishing an equitable stance that critically addresses the shortcomings of reducing objects' or machines' influence to mere human constructs, while also mitigating the excessive attribution of agency to objects or machines. Embracing this ethical perspective involves a shift away from primarily critiquing the political and ethical consequences of algorithms (Beveridge, Figueiredo & Holmes, 2020). Instead, this approach to ethics emphasizes the importance of elucidating and valuing the intricate connections between humans and machines in intricate ecological classroom networks. As students engage with AI-generated texts, their writing strategies evolve, adapting to the possibilities and limitations of the tool. While writing studies has acknowledged this ongoing negotiation with both human and nonhuman components, this dynamic interaction offers a reconceptualization of how practitioners might develop ethical practices as they emerge within classroom settings.

Methodology

Research design

This research is a single-case study focusing on a student's experience with ChatGPT as a writing partner in a Composition II course. We chose the case study approach to gain insights into a "larger class of cases" (Gerring, 2007, p. 4). Initially, the syllabus policy (Appendix A) invited students to use AI TGs within certain parameters and also asked them to document their use. Somewhat surprisingly, only one student followed through with this use and documentation. While more than one student may have used a TG, such use was not definitively clear without documentation and acknowledgement by other students. Upon examining the actual documentation alongside the student's writing, however, it became clear that a single case would provide ample data for exploratory, hypothesis-generating research well-suited to an emerging topic of research such as AI, composition, and ethics. Case studies are common in writing studies research as they help organize data and provide detailed narratives (Gallagher, 2019). Our study aligns with Gallagher's (2019) framework, situated within an online, asynchronous writing course where the student engaged with AI (ChatGPT) and other internet resources to meet course objectives.

Course context

An intensive summer course, English 202 - Composition II was offered as a 5-week, asynchronous class themed around "Questions of Morality in the Vocational University" and asked students to engage with problematic issues in their field of study or future profession. Beyond the thematic focus, the course provided a process-based introduction to secondary research and adopted the *Bedford Researcher* (Palmquist, 2018) as a textbook. Major assignments included a research project proposal, evaluative annotated bibliography, rough draft, peer review, and final draft of the culminating research paper. The second in a sequence of two required composition courses, English 202 represents a fairly typical writing course with the exception, perhaps, of the policy on AI text generators, which was put in place for the first time in summer 2023. The policy permitted the use of AI text generator tools such as ChatGPT under certain conditions:

First, AI tools should not be used as a substitute for your own writing process, but rather, as an additional resource to help you be the best writer possible. While ChatGPT and other AI text tools can help with a number of writing tasks, they should only be used in a way that informs 1) your learning, and 2) your engagement in the processes of research-based writing. Never simply copy/paste an output from ChatGPT and attempt to pass it off as your own. Doing so will result in a failing grade for the assignment and possibly a referral to the Provost's Office for violation of academic integrity....Any time you use ChatGPT for any assignment in this course, (formal or informal), please acknowledge that use in your writing with a statement such as 'The writing of this paper was informed by ChatGPT.' Please also include a screenshot of the prompt and output you consulted. (Course Syllabus, "AI Tools," p. 4).

The emphasis in the policy is on 1) learning and engagement, 2) avoiding copy/paste (or attempting to pass off ChatGPT outputs as original work, and 3) documenting use. The policy also provides four examples of appropriate uses, such as outlining, brainstorming, proofreading, and revising (Appendix A).

Methods and datasets

After the course ended and final grades were posted, the co-author, who was also the course instructor, contacted the student to inquire about their interest in participating in an interview regarding their use of ChatGPT. Receiving the student's consent, an interview protocol was developed and received IRB approval (#23-120-IUP). The student was assured that his decision to participate would not affect his academic standing. The interview, serving as the primary dataset, was conducted as a subsequent stimulated recall, encouraging the student to reflect on course activities, assignments, and ChatGPT sessions. Lasting approximately one hour and twenty minutes, the interview consisted of 29 semi-structured questions covering various categories related to AI usage and experience, ethical considerations, academic integrity, and others. Additional datasets for this study included the instructor's observations of the student's interactions with ChatGPT during class discussions and exercises, the student's final paper, and ChatGPT logs (7 sessions that consist of over 100 individual conversations with ChatGPT). These additional datasets provided supplementary materials for understanding the student's interview responses.

Participant

Ethan (pseudonym) is a 19 year old, white male majoring in Computer Science and minoring in Cyber Security. At the time of the interview, Ethan had completed two regular academic terms and a summer term, meaning he was entering his second year of undergraduate study. While the composition course represented the first time he actually experimented using ChatGPT, Ethan was familiar with the software and generative AI more broadly, having talked with friends about it as well as read about it in online forums such as Reddit. He also admitted that although he had never used ChatGPT, he “knew people that had used it” and “it was something that [he] was interested in” (Interview, 5:08). From the instructor’s perspective, Ethan was academically successful and motivated. His work in the course was above-average, and he had bypassed the required first year composition with a high school AP English credit. Finally, Ethan used the free version of OpenAI’s ChatGPT (GPT 3.5) rather than GPT 4.0 which, at the time of writing, was only available via paid subscription (OpenAI 2023).

Data analysis

Analysis of the interview data followed a systematic six-step process and involved all members of the research group to best extract meaningful and valid conclusions via thematic analysis (Boyatzis, 1998).

1. Transcription: We used Zoom’s automatic transcription tool for generating the initial interview transcript.
2. Familiarization: All members of the research group read and re-read the transcript.
3. Initial coding: Collaborative initial coding was done in a shared Google Doc. Analysis was conducted using the free and open-source software Taguette (taguette.org).
4. Thematic analysis: We analyzed the data through open coding, axial coding, and selective coding using Taguette.
5. Coding and theme refinement: We refined the codes and themes based on our theoretical framework for analyzing rhetorical ecologies and localizing systems of ethics.
6. Selecting salient quotations: Finally, we highlighted key quotations to report in the article.

Study limitation

While this study provides valuable insights into the local ethics surrounding AI usage and policies, it is crucial to acknowledge and address the limitations associated with our research design. The primary limitation stems from the use of a single case study, focusing on the experience of a white, male undergraduate student. We recognize that the choice of a singular case may restrict the generalizability of our findings to a broader population. We call upon future researchers to expand upon our findings by including a more diverse participant pool, which would contribute to a more comprehensive understanding of AI ethics across diverse student populations and various demographics.

To mitigate the limitation and enhance the validity of this study, we have triangulated and cross-verified multiple data sources to gain a deeper insight into the student’s experience by collecting multifaceted data related to the student’s interaction with ChatGPT. In this study, various sources of data have been documented and analyzed, including the student’s interactions with AI through documented ChatGPT logs (7 sessions that consist of over 100 individual conversations with ChatGPT), the in-depth student interview conducted as a subsequent stimulated recall, observations of the students’ interactions with ChatGPT through class discussions and

Table 1
Ethical elements and heuristics for composing with AI.

Ethical Element	Definition	Heuristics for policy and discussion
Pedagogy	The relationship, relationality, and negotiation of meaning between student and teacher	How might the use of generative AI by students or teachers affect their relationship? What sorts of cultures or atmosphere are produced by syllabus policies that address (or fail to address) AI? How does generative AI impact the relations (among students, teachers, texts, policies, and learning outcomes) crucial to learning?
Criticality	Students and teachers’ capacity for and practice of critical thinking and questioning directed at generative AI technologies	What kinds of critiques need to be directed at generative AI from the standpoint of the teacher? What about from the standpoint of the student? How should students practice critical thinking when using AI as a writing partner?
Agency	The dynamic distribution of intent and activity between student writers, AI technologies, elements of classroom ecology, etc.	How should students understand their actions as distributed among technological agents and other elements of the classroom ecology? What can teachers do to help students retain a sense of agency in their writing while using AI TG’s?
Reliability	The social, technical, and material understanding of generative AI technologies capabilities and inabilities	What are the specific affordances and constraints of generative AI tools? How should students learn when to trust and distrust AI tools?
*Accessibility	Access to and capacity for effective use of AI. Accessibility encapsulates both material and economic access to technology as well as social and cognitive access to skills and uses	How should we define access to AI in terms of its material, social, economic, and cognitive implications? Who has access to AI and who doesn’t? How might AI applications act as assistive learning technologies?

exercises, as well as the student's final paper, course assignments, and course syllabus. The interview provided insights from the student participants' perspectives; the classroom observations offered a direct view of the student's class actions and interactions; and the student's final paper and ChatGPT logs served as tangible products of his AI-assisted writing. The act of triangulating data sources has served to strengthen the overall contribution of our study to the field.

Ethical elements in composing with AI

Analysis of the interview transcript led to the negotiated production of a local ethic of the classroom ecology wherein the student-participant Ethan verbalized values, concerns, and emergent conventions around their use of AI, as well as how AI might be integrated in other contexts. Most notable among these are four themes, all of which demonstrate the dynamic ethical negotiation at play in the classroom context. Building on this qualitative data allowed for the creation of a foundational framework and heuristic model capable of informing localized classroom policy, discussion, and exploration for composing with AI (Table 1).

Pedagogy

If we understand the classroom ecology as consisting of the student, teacher, course material and context, and the use of ChatGPT, among other objects, Ethan's construction of a local ethic for using AI was contingent on negotiation and student-teacher relationality, which we viewed as essential to our understanding of pedagogy's etymological makeup: *pais* ("of a child") and *agogos* ("leader"). While this is partly informed by the syllabus policy, which asks students to document their use of any TG (including providing documentation of inputs, or prompts, and outputs), the interview also revealed additional instances of negotiation between the student and instructor in the rhetorical ecology (Rivers & Weber, 2011) of AI usage. Ethan's doubts concerning appropriate and inappropriate usage of AI are illustrated in his discussion of the gray area that exists beyond simply copying and pasting outputs (the latter he understood as definitively plagiarism):

So I wanted to make sure with [the instructor]. Is it okay? If I used this structure, basically because I didn't come up with it. This AI did. I can ask it to revise it for me, and I'm not gonna take the exact output it gives me, but it will come up with ideas like, maybe you could add a sentence here to string these two ideas together, and if you do that it's like, well, you didn't necessarily have that thought yourself. That was a product of this AI. Is it in the same realm as if you just took the output itself and tried to pass it off as your own? That's something I don't know where you draw the line. As to where you know this is plagiarism, or you work with it, you got ideas from it, but as long as the work is yours, then that's okay. (Interview, 41:23)

When asked about his understanding of the syllabus policy, Ethan described the documentation process as a "paper trail" to provide evidence of the full chat logs. During the course, Ethan provided seven such logs to document his AI use, beginning with one in which he used ChatGPT to provide an outline of his research paper. "I think the first time I emailed about ChatGPT was after I had used it to come up with my outline....No one, I think, ever really worries about plagiarizing an outline to an essay. At least, that's not something I was ever aware of. So I wanted to make sure with [the instructor]...Is it OK?" (41:05).

Even while actively negotiating the ethical perimeters of Chat GPT with the instructor, Ethan is well aware that other students were probably avoiding that approach completely. The negotiation between Ethan and the instructor implied he was one of the "responsible users," which, given the capabilities of ChatGPT, might put him in the minority:

I feel like there's always gonna be students that are gonna bend the rules, and...break the rules regardless. So I don't think...even whatever you have in writing, you could have a whole constitution writing for the policy unless you, you know, really crack down on it. I don't think it'll ever be enough to enforce responsible users. You have to assume the goodwill of the people in the class, I guess, and you know...there are people that are gonna break those rules. (1:06)

Ethan's understanding of AI ethics evolved throughout the process of negotiating meaning with his instructor. Ethan shared his concerns over getting inspiration from ChatGPT. Even though he was aware that it was okay to use ChatGPT for generating outlines and brainstorming ideas for his essay, he nevertheless cast doubt on whether it was ethical to do so, because, in his own words, "you didn't necessarily have that thought yourself." Through negotiating with his instructor, it gradually became clear to him that "you got ideas from that, but as long as the work is yours, then that's okay" (42:08).

Student criticality

Elbow (1983) argues that writing teachers should focus on teaching thinking by teaching writing. He believes that the concept of teaching writing is to allow students to think critically about the knowledge that they get either from the readings that are assigned to them or from the teachers. Since a huge part of writing is to engage critically with the literature, the [MLA-CCCC Joint Task Force \(2023\)](#) states that when students fully depend on ChatGPT when summarizing an article, "student reading practices can become uncritically automated in ways that do not aid critical writing instruction or faculty assessment approaches" (p. 6). As a result, writing teachers could lose sight of the cognitive and imaginative aspects of teaching writing, especially its assistance with critical reading comprehension, if their students become fully dependent on any information that AI could generate. In this study, Ethan did not (fully) depend on the information that he received from ChatGPT as he stated:

It'll come up with something, but you know it can't cite sources. It can't tell you. Hey? I got that from this article. You know, by this publisher. Here's the website. So you still have to go back and make sure. Okay, not only did this happen, but every single detail that you just told me actually happened, so that you know, that made me scared. I was really worried that it would give me something that was a really good example. But then you go down a rabbit hole when it's like, oh, this either didn't happen at all, or it happened but...you have details in here that I can't find any credible sources on. (23:34)

Concerns about the potential loss of student criticality in engaging these generative AI tools (Alafnan et al., 2023; Fyfe, 2023; MLA-CCCC, 2023) are valid, yet Ethan's case demonstrates that students can still practice criticality if guided to do so. Ethan did not automatically believe in the sources or the information provided by ChatGPT; rather, he attempted to carry out his own research to find credible resources for the information provided by the AI. Such a process is essential for maintaining student criticality in the face of generative AI. Students who recognize and engage with the limitations of ChatGPT or other TG's will potentially understand that AI can't substitute the full value of writing to enhance one's critical reading strategies.

Agency

Ethan both acknowledges and dismisses ChatGPT as a co-agent during the interview. He welcomes writing as an ecological landscape between people and objects when referencing his own usage of ChatGPT, while romanticizing the autonomous writer when referencing beginning writers. When it comes to authorship, Ethan recognizes ChatGPT as a co-author: "I would definitely say I have to assign partial authorship to GPT, even though I didn't use the exact output it gave me. It did define a lot of the general structure of my whole paper, and that's obviously a huge part of any writing task, you know." (50:08)

By assigning partial authorship to ChatGPT, Ethan both admits and valorizes the distributed nature of agency in an AI-enhanced writing ecology. In other words, AI has played a significant role in generating content, as both human and machine inputs are integral to the creation of the written text. This authorship attribution demonstrates the student's recognition of the AI TG's positive role as a collaborator, highlighting the importance of its contributions while not diminishing the student's involvement. The student's understanding of agency and authorship further reflects the fair position of virtue ethics (Colton & Holmes, 2018; Latour, 2004), an ethical standpoint that provides an opportunity to address the vice of assigning objects' or machines' influence solely to human perceptions, as well as the issue of granting excessive authority to objects or machines.

The distribution of Ethan's agency is further emphasized when he personifies ChatGPT in multiple answers, especially when describing its benefits. He does this when attempting to rationalize its usage for students in a composition class, suggesting that you can "pick it up and treat it as if you were talking to a friend" (58:18), or "use it as your teacher or teacher's assistant...someone you can go to and say 'hey, you know, can you help me sort of reword this, or get my point across clear[?]' (54:57). Such personification is used to explain how ChatGPT provides a "net positive" (59:36) in his view: "it's really like just always having someone, or I guess some thing that I could go to and say like, 'Hey, what's your thoughts on this?'" (59:36). This rhetorical choice helps Ethan recognize ChatGPT's ease of use and accessibility, while rationalizing his usage. If ChatGPT is helpful like a writing professor or TA, then why wouldn't it be ethical?

Despite the various benefits of ChatGPT as a co-agent of writing, Ethan also discusses its limitations to help first-year composition students, those who haven't worked on or fully understood typical rhetorical and writing choices of a first-year writer. He compares this to a math student jumping to use a calculator before fully understanding the math equations without the help of the technological tool.

It could definitely be detrimental [at the first-year level], because again, it'd be like the calculator analogy. You learn how to do it. You know everything on the calculator, but you can't do it on [your own], you know. You can't do it by hand at all...It's sort of that same sort of thing, I think. Where, if you jump on it too early, then you don't have those skills yourself. And so it sort of becomes a crutch for you. (1:00)

Even though Ethan assigned partial agency to ChatGPT, he didn't fully embrace the ecological and networked understanding of agency and authorship. At multiple points he voices a binary between "learning how to write" and establishing "those skills yourself" versus using ChatGPT to enhance your writing process to have a "net positive impact." As discussed previously, scholars in writing studies have increasingly problematized the romantic understanding of an autonomous learner or writer, especially with the advent of digital communication technologies (Cooper, 2019; Gries, 2015; Jiang, 2019; Jiang & Tham, 2022; Jordan, 2015; Micciche, 2014). AI authorship further, and perhaps more dramatically than before, blurs the boundary of human and nonhuman actors that co-produce writing in the digital rhetorical ecology. Ethan's perspective on this binary, however, assumes that deeper learning for beginner writers happens alone, and that ChatGPT is somehow preventing a person from building an intrinsic set of writing skills, rather than a dependency on a machine. In a way, Ethan's remarks serve to counter this idea that learning happens somewhere *in between* our sense of agency and other objects outside of ourselves.

While Ethan expressed doubt around certain uses of ChatGPT, he made clear that there were also distinct boundaries pertaining to academic integrity as it relates to the originality and authenticity of his own composition or writing. His discussion of plagiarism, for instance, demonstrates the value he placed in doing original work: "I think everyone understands, like, if you just take the output generated by [ChatGPT] and then try to pass it off as your own, obviously, that's wrong. That's plagiarism." (40:46). When asked about his writing processes after receiving input from ChatGPT, for instance, Ethan recounts the following:

I sort of went back through my essay. First, I added a lot of sign posting stuff manually to certain paragraphs, and then that's where I decided to change that paragraph I talked about with ChatGPT and the final discussion. That's where I decided to sort of

change that one into an Intro paragraph, and then I came up with two additional paragraphs. I think one or two additional paragraphs after that, just sort of on my own. I didn't put them into GPT at all, because again, it just didn't seem like it would be worth it at that point.

Here, Ethan shared his "manual" editing practices beyond AI-generated texts that further allowed him to circumvent the ethical concern regarding plagiarism. Editing and revising the content generated by ChatGPT also demonstrated that the student didn't grant excessive authority to ChatGPT.

Reliability

The ethical values surrounding Ethan's perception of ChatGPT's reliability were expressed through his discussion of trust and distrust. As became obvious not only in his consistent and thorough use of ChatGPT, Ethan trusted the software enough to use it for certain processes, especially those related to outlining, brainstorming, and revising (which were also prompted by policy language drafted by the instructor co-author and included in the course syllabus).

As far as, like, the revisions it would make to my actual paragraphs...I didn't see any reason not to trust that. You know – the revisions it was making were optimal. I just sort of accepted, like, okay, if it's going to revise this, then what it's giving me is... good. It's truly going to benefit me to consider the revisions it makes. I'm not going to have to worry about it trying to, like, sabotage me, or anything like that, as far as changing my wording or sentence structures. (38:16)

Other aspects or processes could *not* be fully trusted to ChatGPT – specifically what Ethan calls "real-life examples" and citations in MLA style. However, such distrust did not mean that Ethan limited his use of ChatGPT for brainstorming examples or even for checking documentation style. Rather, he would allow ChatGPT to come up with examples for his paper, and then double-check using Google to make sure those examples were credible. For issues related to MLA, additionally, Ethan discussed using the Purdue OWL website to make sure that what ChatGPT was providing was accurate.

Distrust I really had with it was regarding the examples and also the MLA citations. I didn't know if those would be completely accurate, since it's not actually crawling the web. Real life examples: there is very little trust for that. I didn't take anything it said there for, for you know, for word, I would always go back to Google and just do a classic search and see, okay, did this event happen if it did find a couple of articles and make sure that every detail ChatGPT brought up was mentioned in at least one of these articles somewhere by a credible source. (38:52)

What's notable in this excerpt is that Ethan's perspective on the reliability of ChatGPT is extremely fluid, dependent on not only the type of task, but also on his own intuition and understanding of what a TG can accomplish paired with an understanding of his own writing skill and capacity. First, Ethan understands that ChatGPT is more of a probability engine than an authority on the topic. Second, while he admits to not being a "great writer," he also recognizes that he writes at a "decent enough level" and that his own evaluation of ChatGPT outputs provides the ultimate assessment of its reliability:

I can at least write at, you know, a decent enough level that my work isn't going to get completely canned when I turn it in. So... you still have to be able to do it on your own and to look at the output it gives you and say, Okay. What's good about that? What can I actually use? And, you know, try to [figure out] how to fit all of that together. (35:34)

The issue of trust is nuanced because it coexists with its counterpart: distrust. Despite its various advantages, ChatGPT has limitations that have engendered skepticism in Ethan. These limitations include a lack of contextual understanding, the inability to capture nuances, and potential inaccuracies in terms of citations. While ChatGPT can be a powerful tool for remixing corrections, it is not able to generate content that is completely error-free. As Ethan's experience demonstrates, ChatGPT can inadvertently generate inaccurate content and propagate misinformation, especially since it can't cite its own sources and has limitations on the currency of its information (its latest upgrade would allow it to have knowledge up to April 2023). This raises ethical dilemmas and trust issues, as students must remain vigilant and critical about the content ChatGPT generates.

Discussion

This study focused on emergent ethical concerns regarding AI and writing instruction, illustrating the production of a local ethic. The themes (or elements) surfacing within the data reinforce the argument that educators and researchers should follow a more proactive approach when creating ethical guidelines, allowing students to experience and articulate these issues both for themselves and collaboratively with instructors.

In our analysis related to the theme of pedagogy, we learned that Ethan actively sought guidance from the instructor while using ChatGPT, despite the fact that he fully understood the AI policy as articulated in the syllabus. His understanding of AI ethics was therefore enhanced through the process of negotiating meaning with his instructor, reinforcing the notion that rhetorical understanding is emergent and developed through a complex ecology of text, writers and readers (Rivers & Weber, 2011). However, Ethan hesitated to believe his usage of the TG was acceptable under the courses' pedagogical constraints, and regularly reflected on his own possible examples of plagiarism. He negotiated these larger concerns with the instructor and acknowledged that the syllabus guidelines might not be followed by other students. These reflections raised concerns related to our second theme, student criticality, where writing has consistently been defined as a tool to help one explore ideas and think critically (Elbow, 1983). While Ethan recognizes

ChatGPT's benefits for helping him by working as a writing partner or coach, he consistently seemed confused as to whether the ideas AI helped generate were completely his or not, and whether other students would simply abuse AI and circumvent writing the paper all together. Both concerns could become barriers for Ethan to accept the notion that writing leads to critical thinking, since the former questions his own agency as a writer while the latter might lead Ethan, or other students more generally, to feel socially pressured in using ChatGPT in a way that disregards writing as a mode to discovery.

Ethan's assignment of partial authorship to ChatGPT demonstrates his acknowledgement of the distributed nature of his agency. However, he suggests beginner writers should not be using ChatGPT, arguing that only a writer who has the writing skills "yourself" can successfully manage AI without plagiarizing. Ethan suggests these skills help him establish authority over the TG while staying within the perimeters of the syllabus policy. For Ethan, students who are hoping to utilize ChatGPT without plagiarizing must not only have a desire to establish a sense of agency as a writer but also recognize that writers are always participating in a larger ecological landscape.

In terms of reliability, Ethan trusted ChatGPT to make revisions to his work but questioned its ability to generate examples and MLA citations. However, while understanding these limitations, Ethan still searched for brainstorming examples and citations with ChatGPT, checking this content with a skeptical eye. Thus, his perspective on ChatGPT's reliability remained fluid and dependent on the type of task and his own understanding of AI and LLMs. Given Ethan's ability to remain open and creative with his usage of AI, we believe his relationship with TGs will continue to grow while the ethical perimeters he imagines will constantly be negotiated, especially as AI becomes more effective for meaning-making purposes.

While we negotiate ethics within the educational space, larger limitations still exist with both ChatGPT within this study and AI systems in general. Research on TGs is still nascent and writing studies scholars are bound to miscalculate, make faulty assumptions, and generally fail to fully perceive the implications of generative AI's rapid advancements. The accuracy of information continues to be a major issue with ChatGPT with experts suggesting the platform can only augment, not substitute, existing labor (Tayeb, 2023). ChatGPT has also been described as having a left-leaning political bias in its responses (Baum & Villasenor, 2023), and can be easily made to generate toxic, racist responses when assigned a persona (Wiggers, 2023). These issues stem from how ChatGPT curates its training data, or is shaped through reinforcement learning with human feedback (RLHF), which is an attempt to align LLMs with human values (Baum & Villasenor, 2023). Using outsourced and exploited labor to label harmful content has also come under scrutiny (Perrigo, 2023). As long as AI is driven by data capture without caution or consent, these deeper issues will largely remain (Crawford, 2021). Data carries with it individual meaning, personal histories and injustices, which is largely ignored for the sake of machine abstractions and operations (Crawford, 2021). That doesn't mean we should avoid or disengage with this research all together; active exploration and staying curious with these tools is essential, especially within our local spaces.

Implications for AI policy in the composition classroom

The rapid evolution of AI Text Generators (TGs) has prompted transformative changes in the composition classroom, sparking concerns about plagiarism, access, and critical thinking. It's imperative to teach students ethical AI writing tool use, ensuring they function as aids rather than replacements. Instructors must also actively engage in this process. While recognizing the value in creating recommendations for classroom policy, as seen with the MLA/CCCC joint task force and other entities, we urge writing teachers to explore the ethical dimensions of AI within their contexts, fostering the creation of local ethical systems. Our article focuses on locating the ethical elements of AI composition and foundational theory-building for an ethical engagement framework, emerging through active negotiation, practice, and dialogue with a student's interaction with AI. Whether teachers integrate generative AI into their pedagogies or impose limitations, explore ethical parameters, or implement static policies, a local ethic is ever-present. This article advocates for collaborative efforts between teachers and students to formulate a local ethic governing the educational use of tools like ChatGPT. Drawing from study results, we have crafted a model of ethical elements and heuristics for composing with AI, serving as a starting point for localized classroom policies, discussions, and student practices (Table 1). We acknowledge that instructors may not always have the freedom to determine ethical policies and recommend conversations within the context of departmental and institutional guidelines.

The ethical elements explained in Table 1 are borne out of our analysis of the interview data with the exception of the fifth and final element, namely "Accessibility." While there was a small discussion of accessibility, the amount of data didn't warrant a full analysis. However, recognizing its significance and engagement by a number of other scholars (Farrokhnia et al., 2023) we include this item in our final model.

Avenues for future research

Based on the findings, we recommend that teachers and educators attend to the local ethics of negotiation within rhetorical ecologies, which involves prompting dialogues with their students and making those dialogues meaningful in the construction of assignments, activities, and course policy. We also recommend critical attention be paid to the cultivation of students' ecological understanding of AI authorship, agency, reliability, and accessibility which parallels exploring the complex networked ecologies that shape the experiences of human-machine interactions in the composition classroom. We suggest that future research on AI TGs further explore students' interactions with AI TGs in local contexts and add to, revise, and complicate the model we provide. Future studies may delve into learners' experiences with AI in a variety of learning settings, such as second language writing and transnational contexts. In addition to interviews, researchers may also conduct longitudinal studies, such as narrative inquiries or pre- and post-surveys, that will allow scholars to capture the continuity of student writers' evolving understanding of ethics and AI. Longitudinal

research, additionally, may allow researchers and teachers to gain further insight into the ways that students' relations with AI change and evolve over time.

CRedit authorship contribution statement

Matthew A. Vetter: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing. **Brent Lucia:** Conceptualization, Data curation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Jialei Jiang:** Conceptualization, Data curation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Mahmoud Othman:** Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors have no competing interests, financial or otherwise, to report.

Data availability

Data will be made available on request.

Appendix A

AI Tools: The use of AI text generator tools such as ChatGPT is permitted in this course under certain conditions. First, AI tools should not be used as a substitute for your own writing process, but rather, as an additional resource to help you be the best writer possible. While ChatGPT and other AI text tools can help with a number of writing tasks, they should only be used in a way that informs 1) your learning, and 2) your engagement in the processes of research-based writing. Never simply copy/paste an output from ChatGPT and attempt to pass it off as your own. Doing so will result in a failing grade for the assignment and possibly a referral to the Provost's Office for violation of academic integrity. Some useful ways to engage ChatGPT or other AI writing tools:

- 1) As a conversational partner for brainstorming. E.g. List some possible ethical issues in the field of nursing.
- 2) As a writing coach. E.g. Read my draft of this paper so far and identify issues with the sentence structure and argument.
- 3) As a proofreader. E.g. Read this section of my paper looking for grammatical errors.
- 4) As an outliner. E.g. Give me an outline for a research paper based on my tentative research question: What should be done about the problem of racial profiling in police work?

Any time you use ChatGPT for any assignment in this course, (formal or informal), please acknowledge that use in your writing with a statement such as "The writing of this paper was informed by ChatGPT." Please also include a screenshot of the prompt and output you consulted.

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